

Geodesy and Coordinates

GEOG 1026

Department of Geography
National Taiwan University

Principal Triangulation of Britain, 1860, Alexander R. Clarke



Last Week

- Spatial data, geospatial data, geographic data
- Geographic information
- Geographic thinking

Geographic Data

- Data that has a “spatial” component
- Data is connected to a place in the Earth
- Numerical/digital values in GIS represent physical objects
- Represent the location, size, and shape of physical objects
- May include more attributes providing other information
- Describe geographic phenomena related to physical and human geography

Geographic Data

- Data that has a “spatial” component
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- Represent the location, size, and shape of physical objects
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This Week

- Geodesy (測地學、大地測量學)
- Figure of Earth (地球形狀)
- Coordinate system (坐標系統)

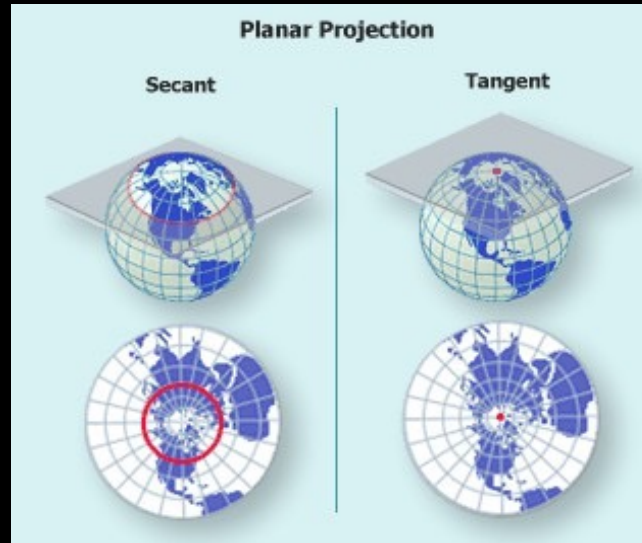
Geodesy (測地學、大地測量學)

- Geo + desy
- The science that determines the figure of the Earth and the inter-relation of selected points on its surface.
- A branch of applied mathematics that determines the size and shape of the Earth and the definition of coordinate systems for the three-dimensional positioning.
- Terrestrial geodesy (大地測地學), satellite geodesy (衛星測地學), image geodesy (影像測地學)

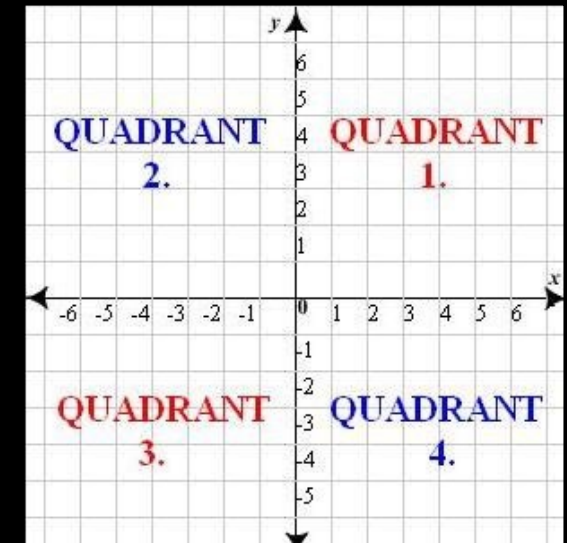
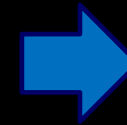
Key Concept



Geographic
Coordinate System



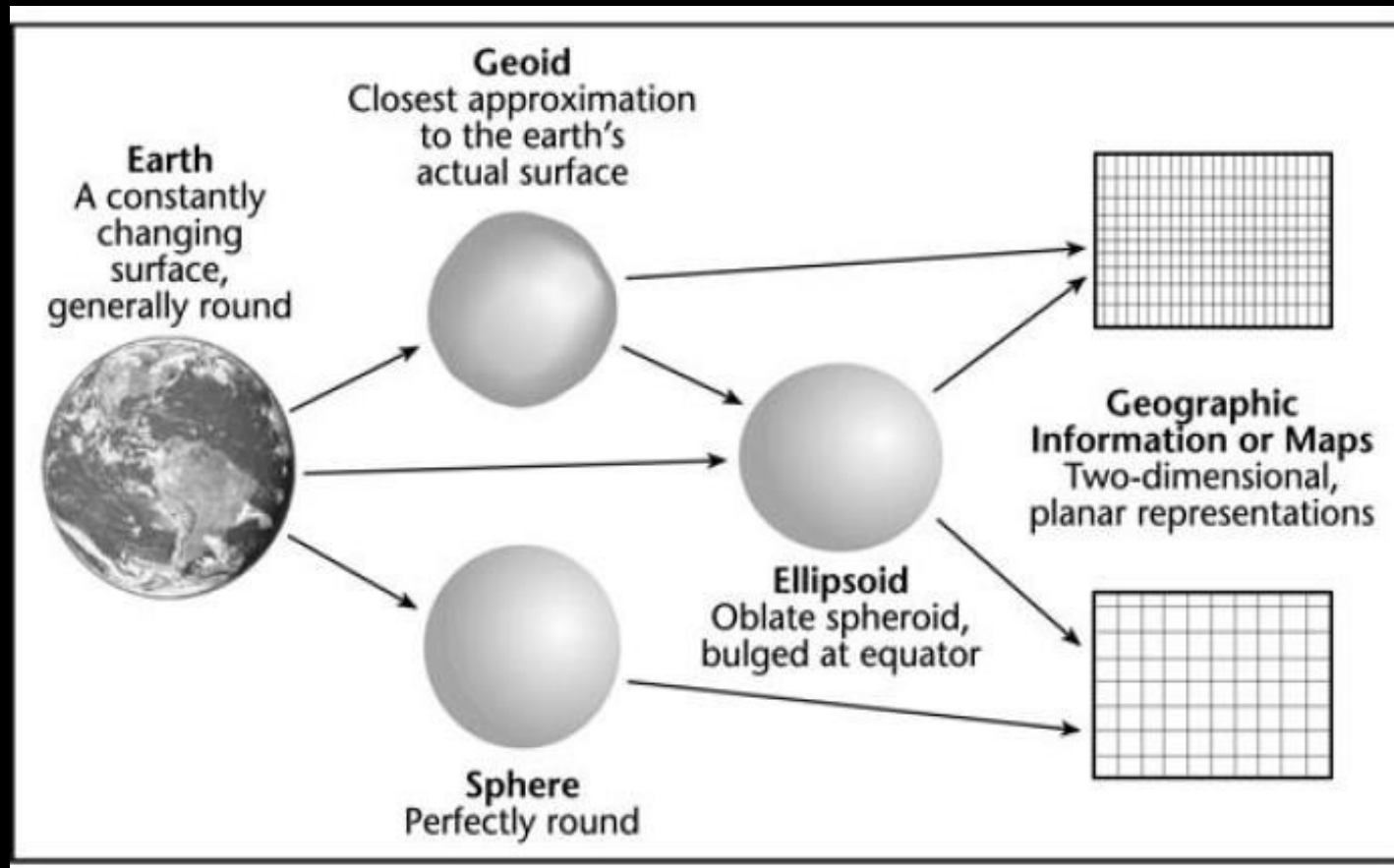
Map
Projection



Plane
Coordinate



Define the Globe: Geographic Coordinates





The Blue Marble, 1972, Harrison Schmitt, NASA



Egyptian



Norse



Hindu



Mayan



Inca



Navajo



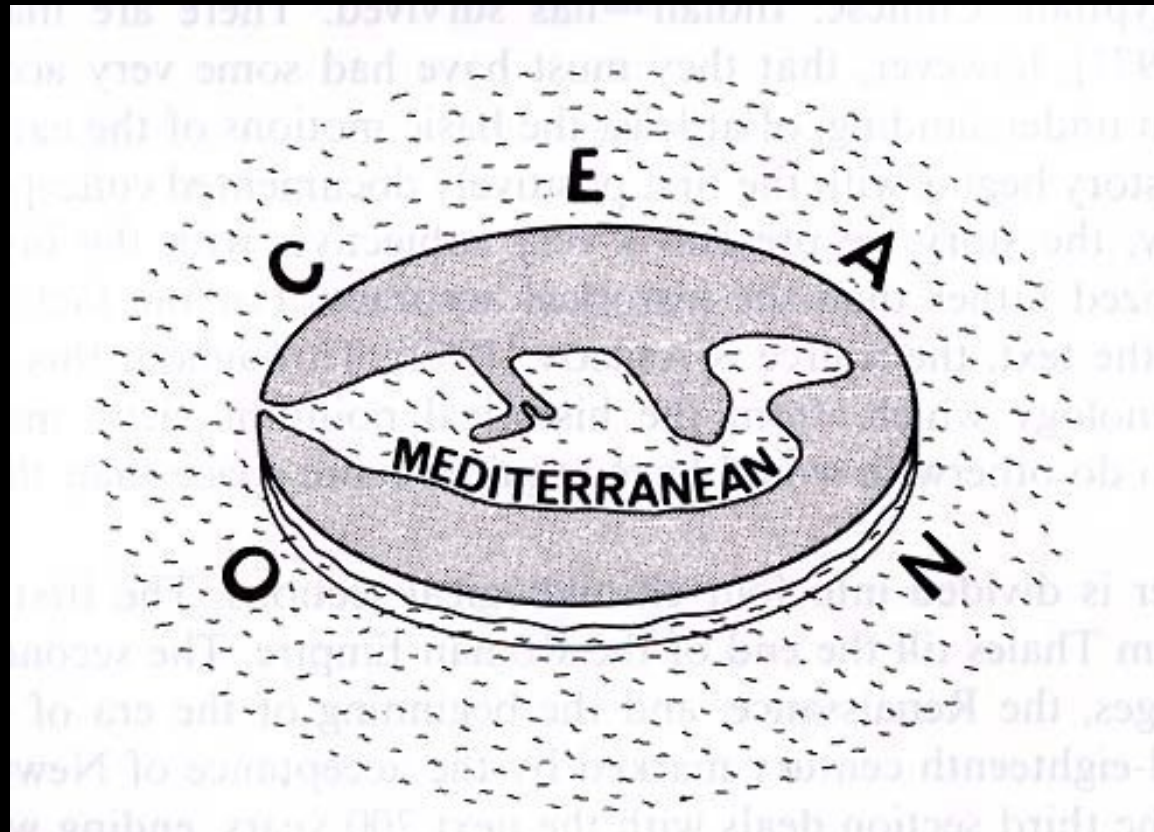
Hebrew



NASA

Earth: A Flat Disk Floating on Water

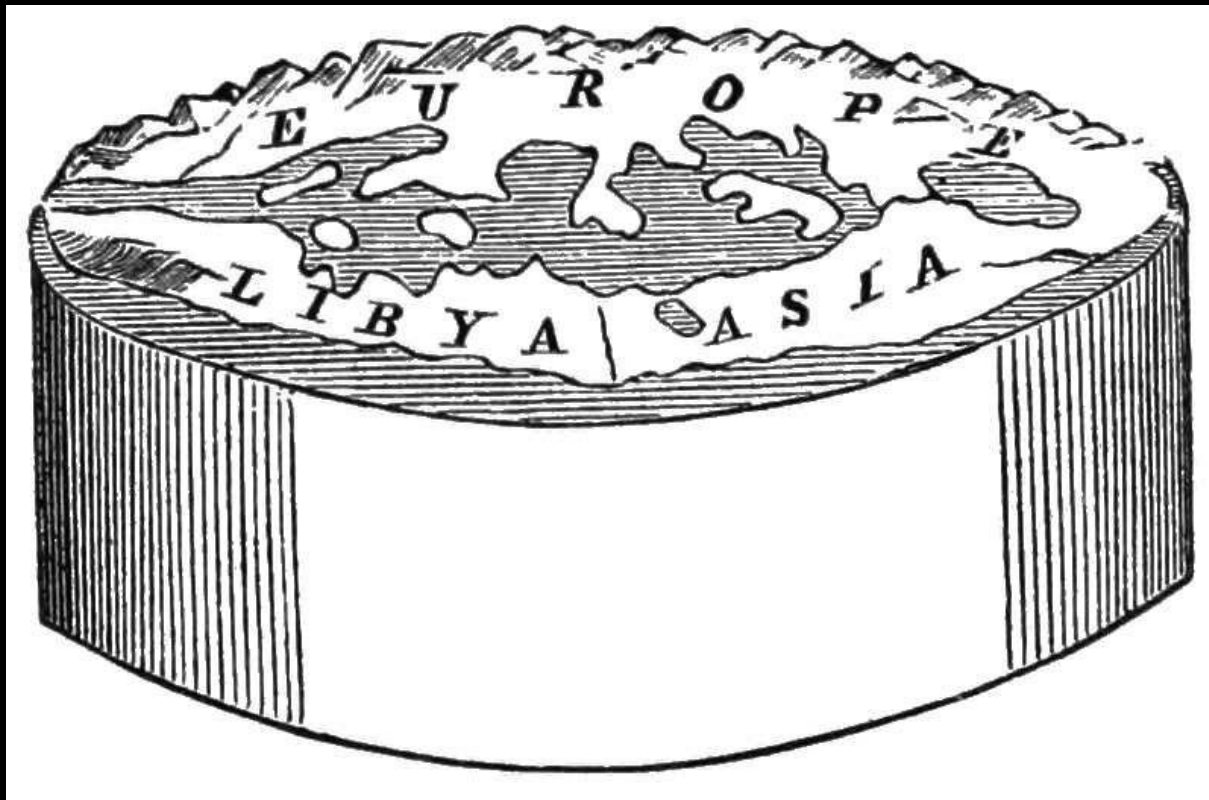
- Thales of Miletus (c. 624 – 547 BC) 古希臘哲學家泰利斯，古希臘七賢之一



Vanicek and Krakiwsky (1986)

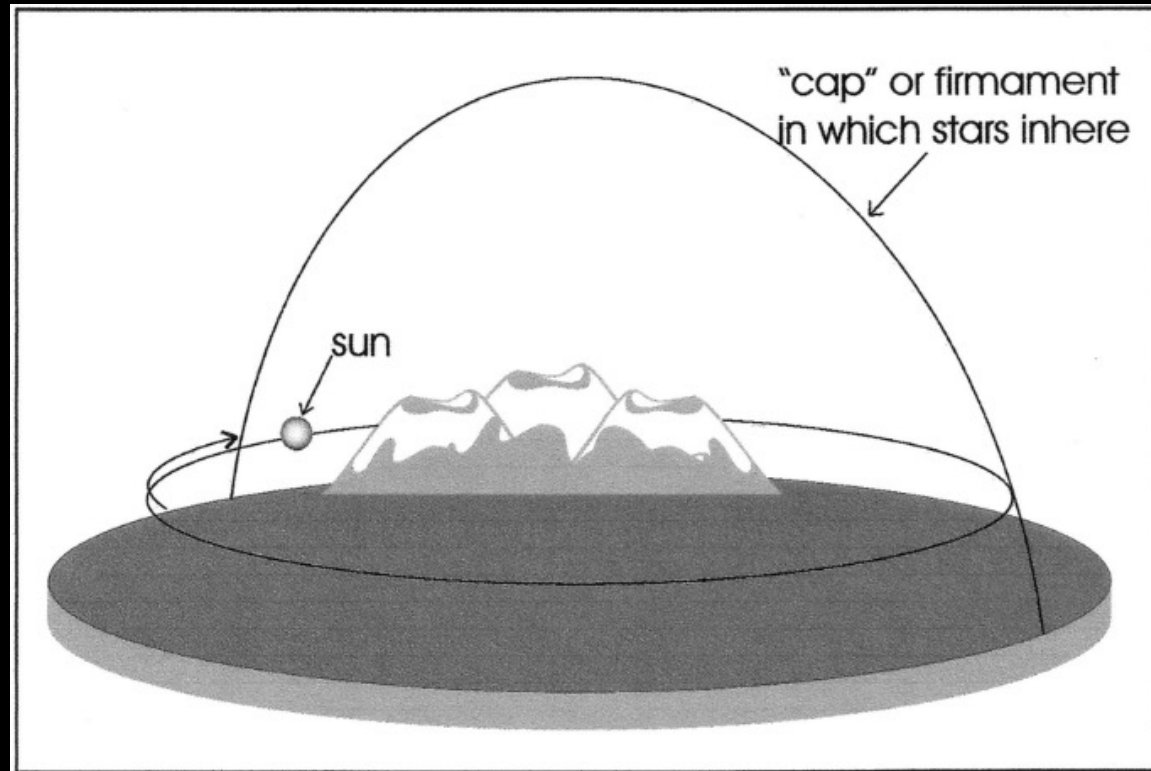
Earth: A Cylindrical Earth

- Anaximander (611 – 546BC) 古希臘哲學家阿那克西曼德



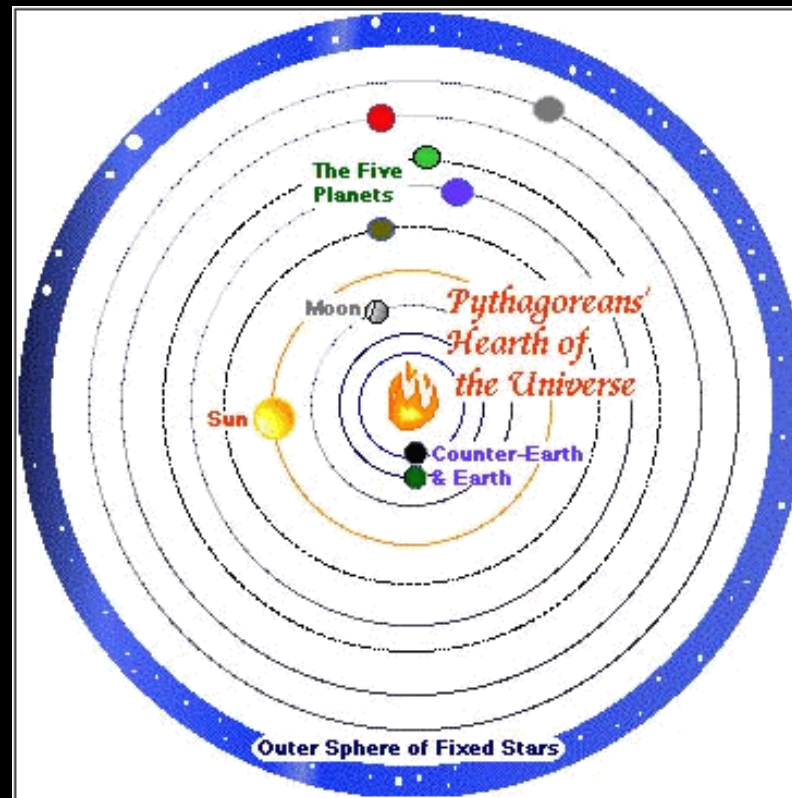
Earth: A Flat Disk Floating on Air Pressure

- Anaximenes of Miletus (c.585 – 525BC) 古希臘哲學家阿那克西美尼



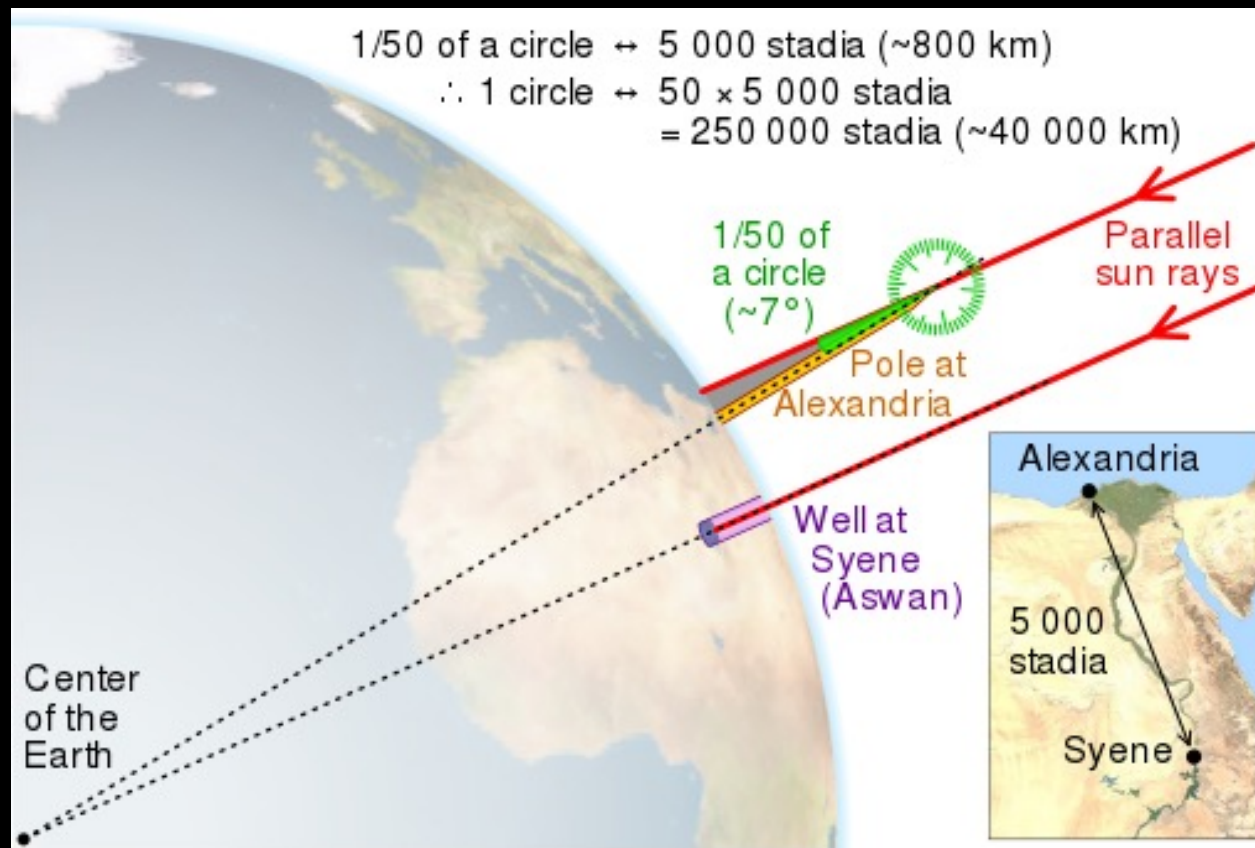
Earth: A Spherical Earth

- Pythagoras (c.570 – 495BC) and Aristotle (384-322 BC)?
- <https://antigonejournal.com/2023/09/ancient-greeks-earth-round/>



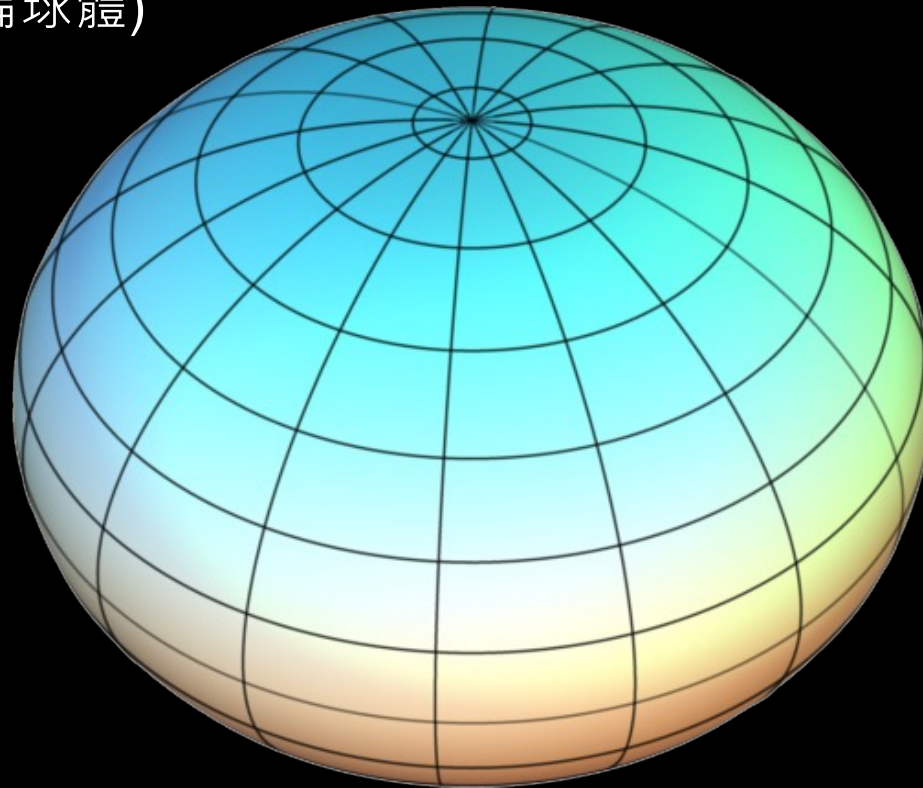
Measuring the Spherical Earth

- Eratosthenes of Cyrene (c.276 – 194BC) 古希臘地理學家埃拉托色尼



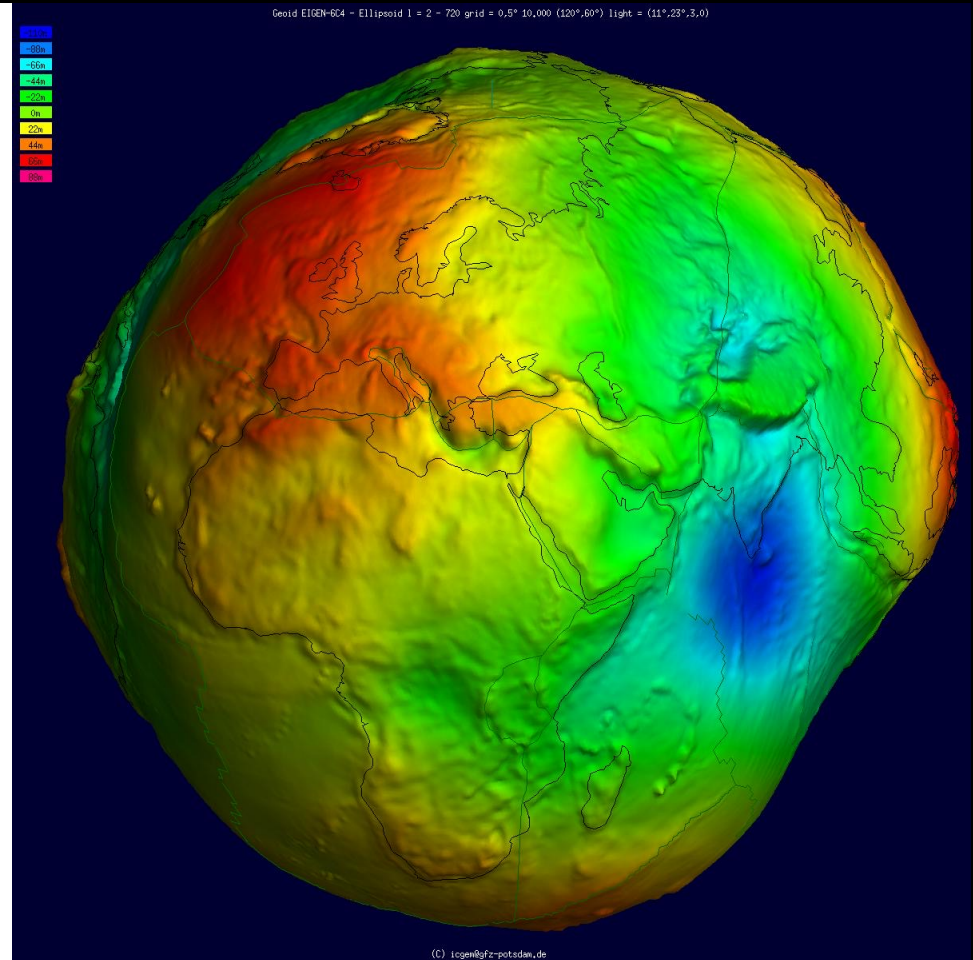
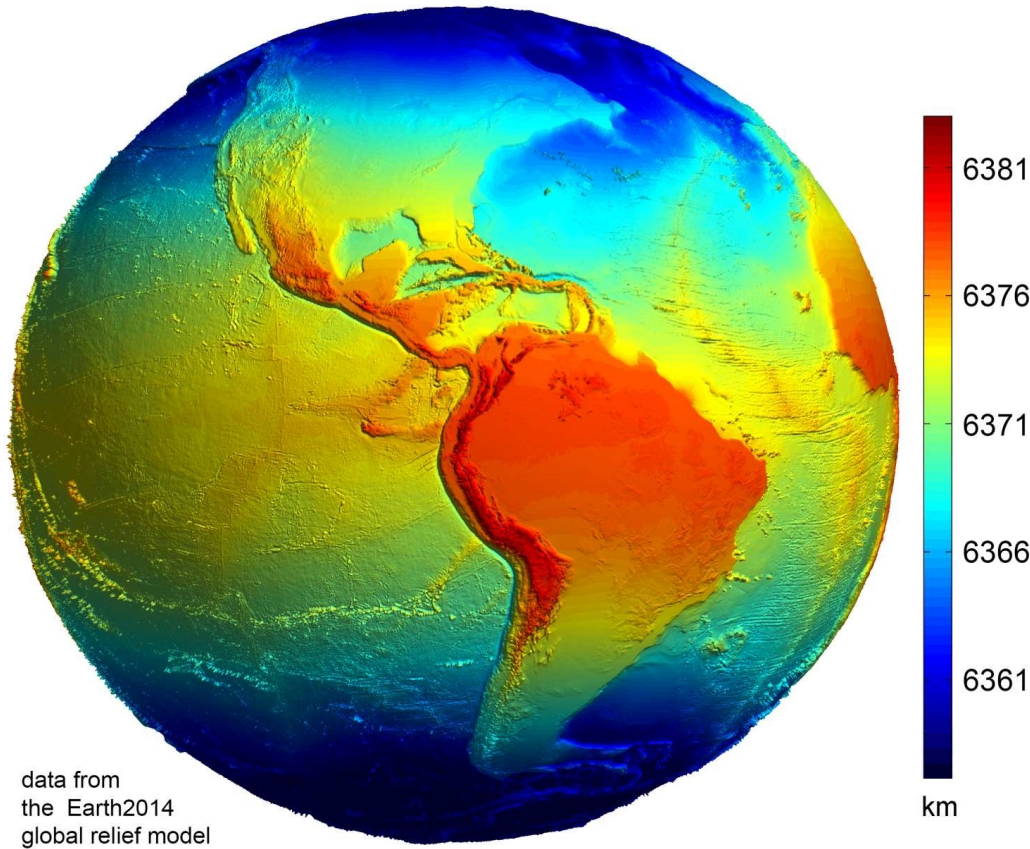
Earth: An Ellipsoid (橢球體)

Oblate spheroid (扁球體)



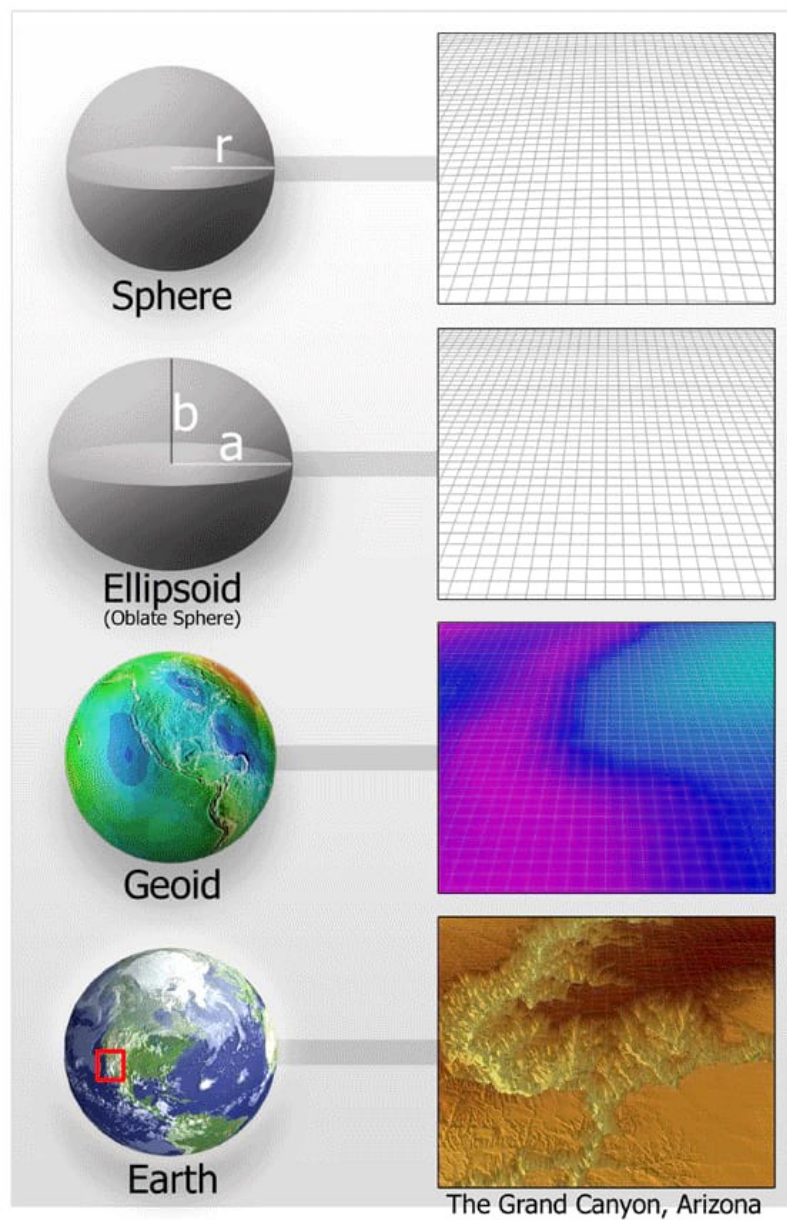
Earth: Geoid (大地水準面)

Shape of the Earth
distances of relief points to the geocentre



Earth: Geoid (大地水準面)

- The geoid is an equal-potential surface defined by the Earth's gravity field.
- It is an irregular surface that cannot be described by a simple math equation.
- The geoid is commonly used as a reference for mean sea-level in elevation measurements.
- A reference ellipsoid may be selected to closely represent the geoid in local areas.

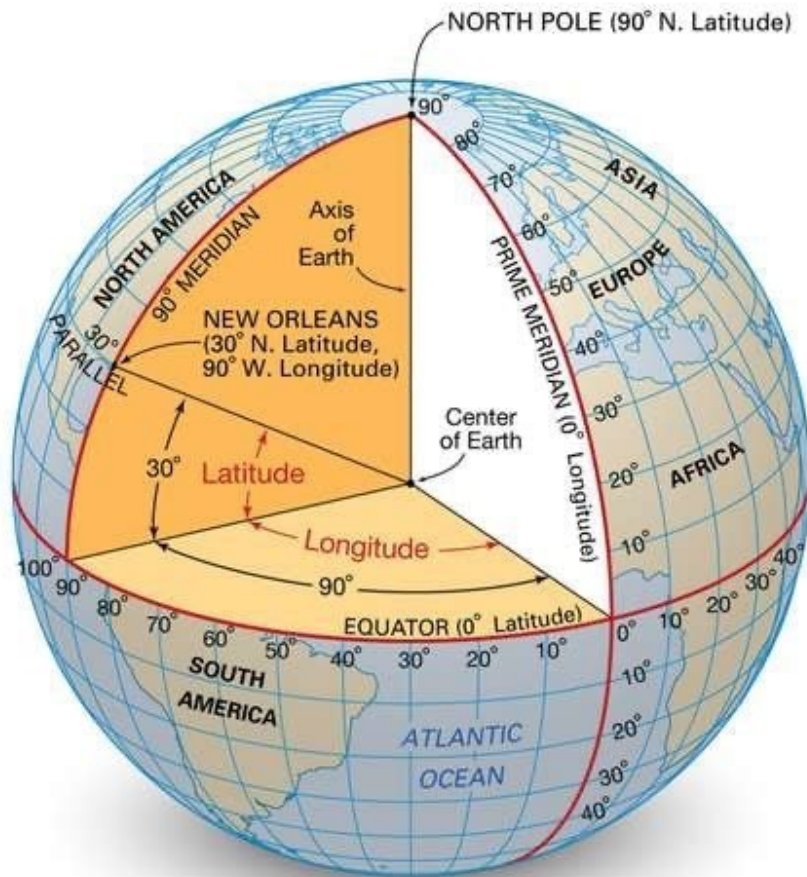


1. Authalic sphere (標準球體)

2. Reference ellipsoid (參考橢球體)

3. Geoid (大地水準面)

1. Authalic Sphere (標準球體)



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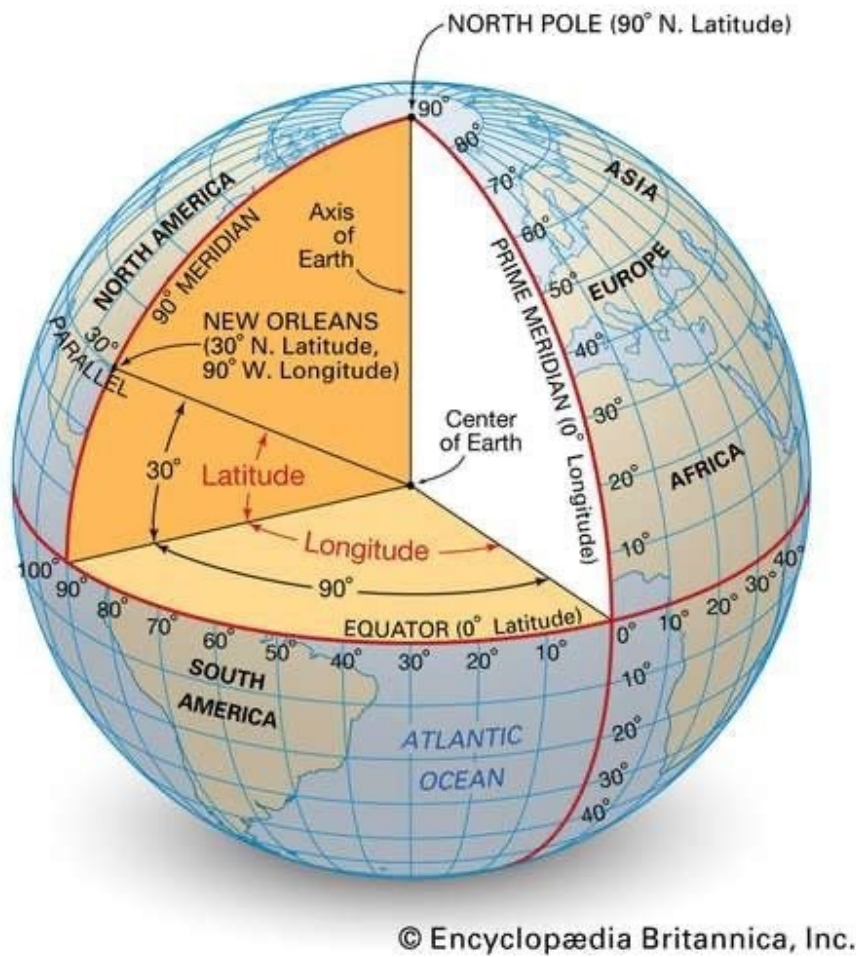
- Unit of latitude and longitude
- DMS format (degree, min, sec)
 - 45° 33' 37" N, 94° 9' 44" W
- Decimal degree
 - -94.162222, 45.560278
- Transformation

若有一經緯度是：a度、b分、c秒，以十進位表示及的公式如下：

$$\text{十進制} = a + \frac{b}{60} + \frac{c}{3600}$$

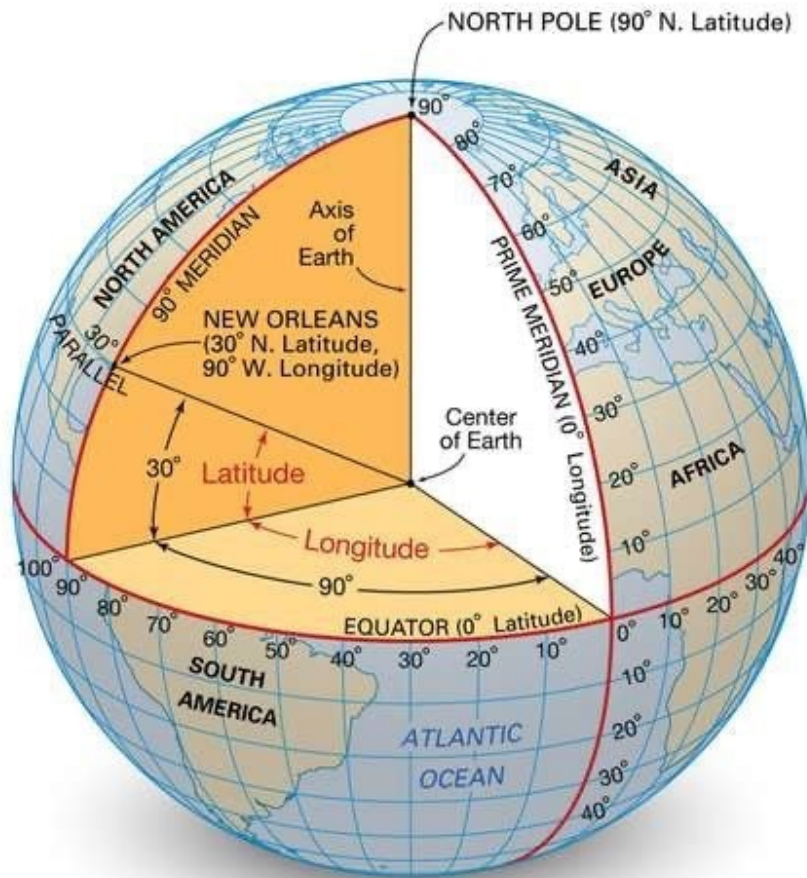
$$\begin{aligned} &121^{\circ}18'32'' \text{ E} \\ &= 121 + 18/60 + 32/3600 \\ &= 121.3089 \end{aligned}$$

1. Authalic Sphere (標準球體)

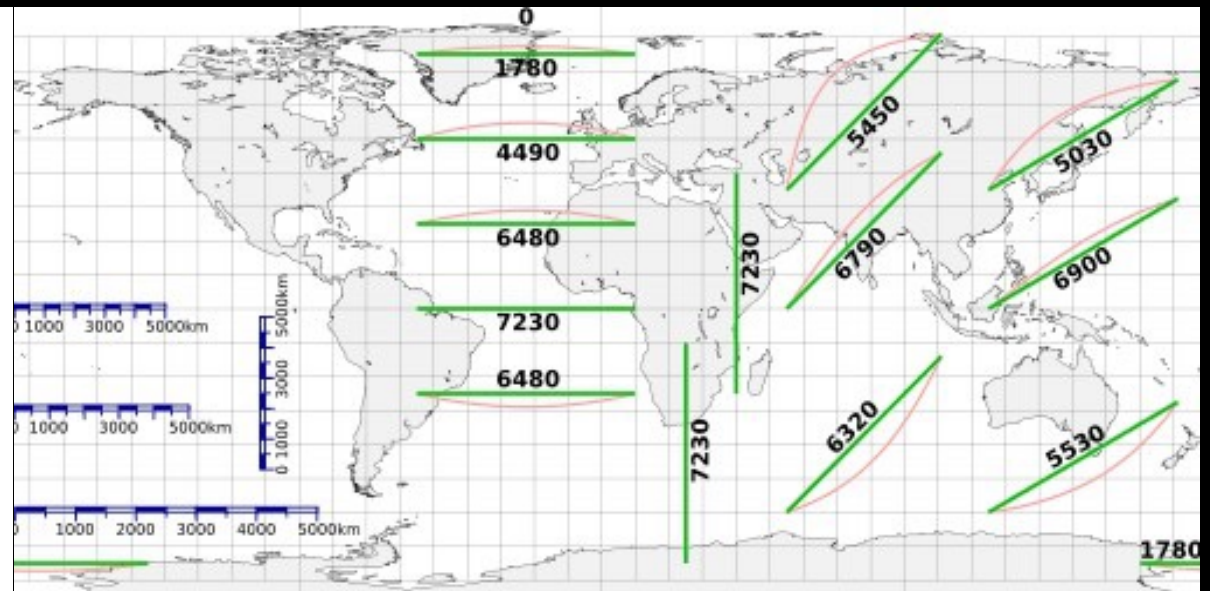


- Longitude (經度)
- Meridian (經線、子午線)
- Prime meridian (本初子午線)
- Latitude (緯度)
- Parallel (緯線)
- North pole (90°N Latitude) (北極點)
- South pole (90°S Latitude) (南極點)
- Equator (0° Latitude) (赤道)
- Origin of geographic coordinate system (地理坐標原點)

Convergence of the Meridians



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- The surface distance between lines of longitude narrows going away from the equator, until all the meridians converge at the poles.

Lenth of the Parallels

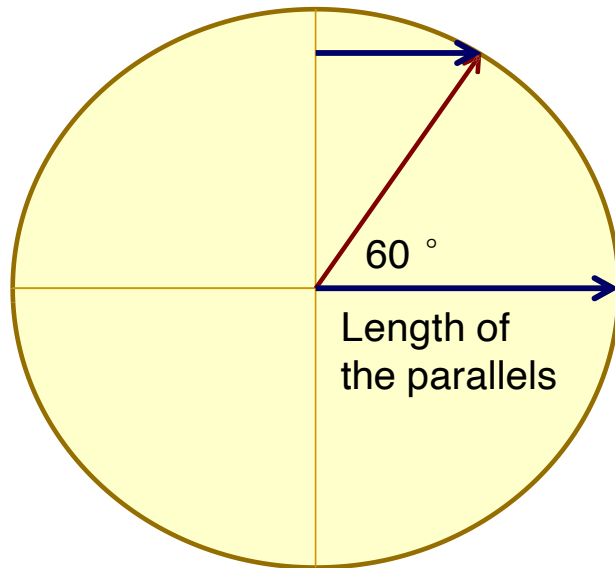
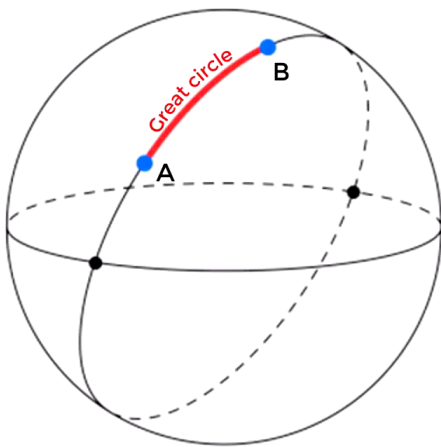
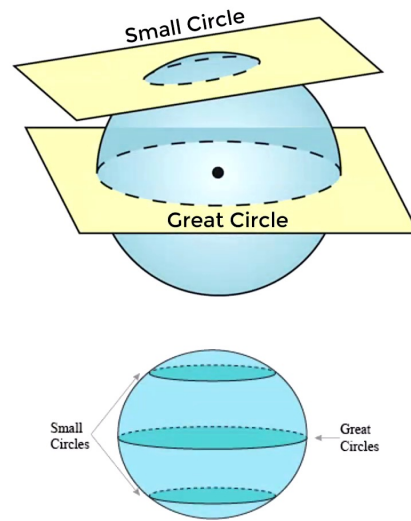
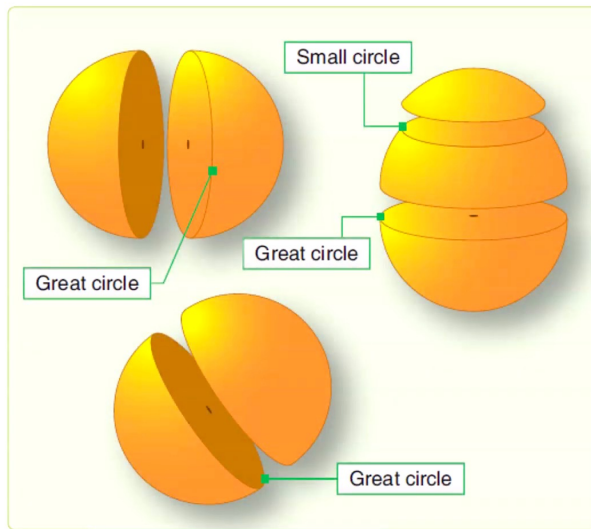


TABLE 2.2 LENGTH OF PARALLELS ON A PERFECT SPHERE ($r = 1.0$) CIRCUMFERENCE (EQUATOR) $= 2\pi r$ (OR πd) $= 6.28318$

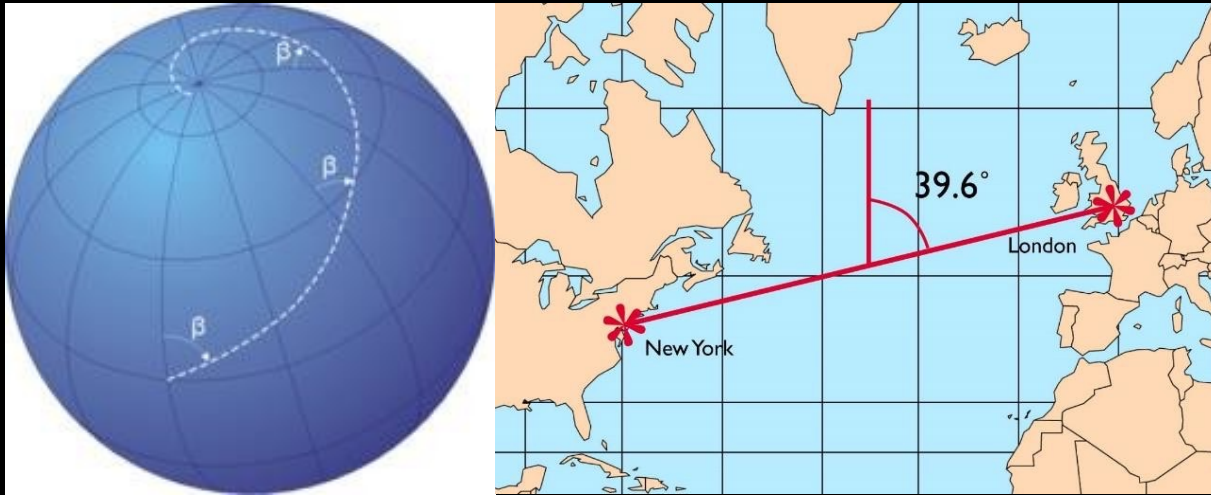
Latitude (°)	Length of Parallel	Percent of Equatorial Length
0 (equator)	6.2831	100.00
5	6.2592	99.61
10	6.1877	98.48
15	6.0690	96.59
20	5.9042	93.96
25	5.6945	90.63
30	5.4414	86.60
35	5.1468	81.91
40	4.8132	76.60
45	4.4428	70.71
50	4.0387	64.27
55	3.6039	57.35
60	3.1416	50.00
65	2.6554	42.26
70	2.1490	34.20
75	1.6262	25.88
80	1.0911	17.36
85	.5476	8.71
90	.0000	.00

Great Circle (大圓)

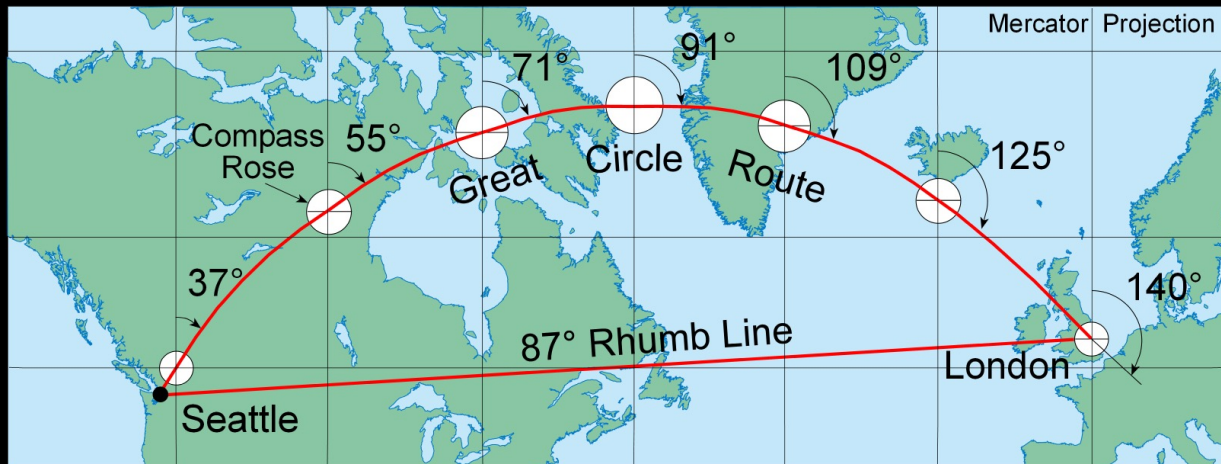


- The plane forming the great circle bisects the spherical surface.
- Great circles always bisect other great circles.
- An arc segment of a great circle is the **shortest distance between two points** on the surface.
- All meridians are great circles, and the equator is also a great circle.

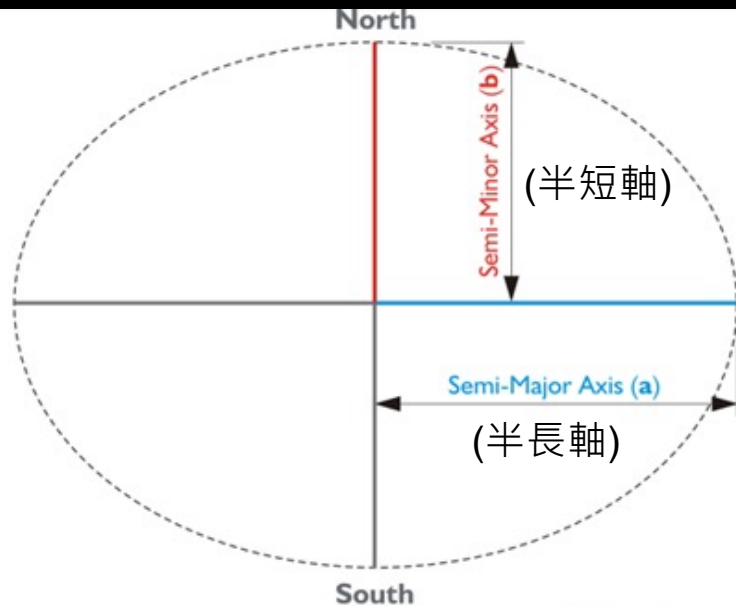
Loxodrome (Rhumb Line) (恆向線)



- Constant azimuth (方位角)
- The same intersection angle (交角) with each meridian



2. Reference Ellipsoid (參考橢球體)



N - S The axis of revolution for generating the ellipsoid

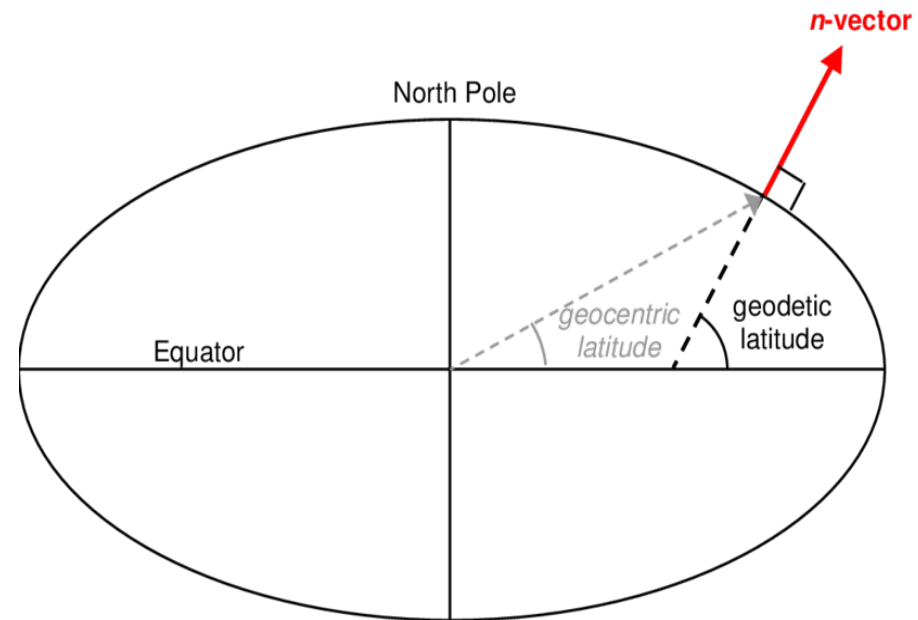
flattening $f = 1 - \frac{b}{a}$

eccentricity $e^2 = 2f - f^2$

a - half of the major axis
the semi-major axis

b - half of the minor axis
the semi-minor axis

Parameters of a Biaxial Ellipsoid

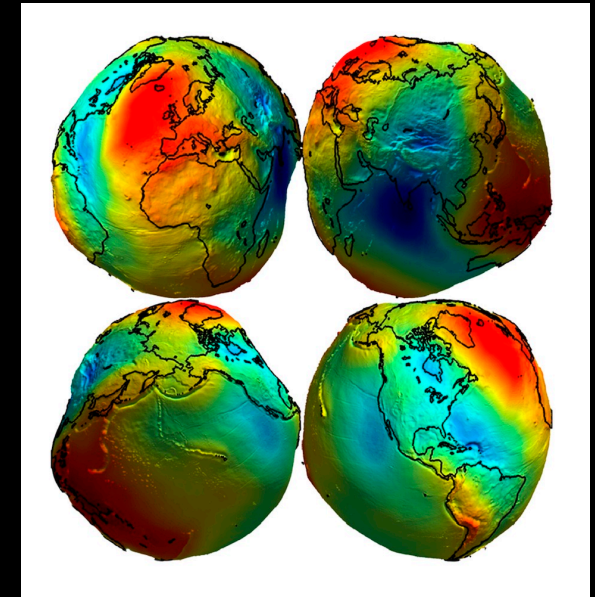
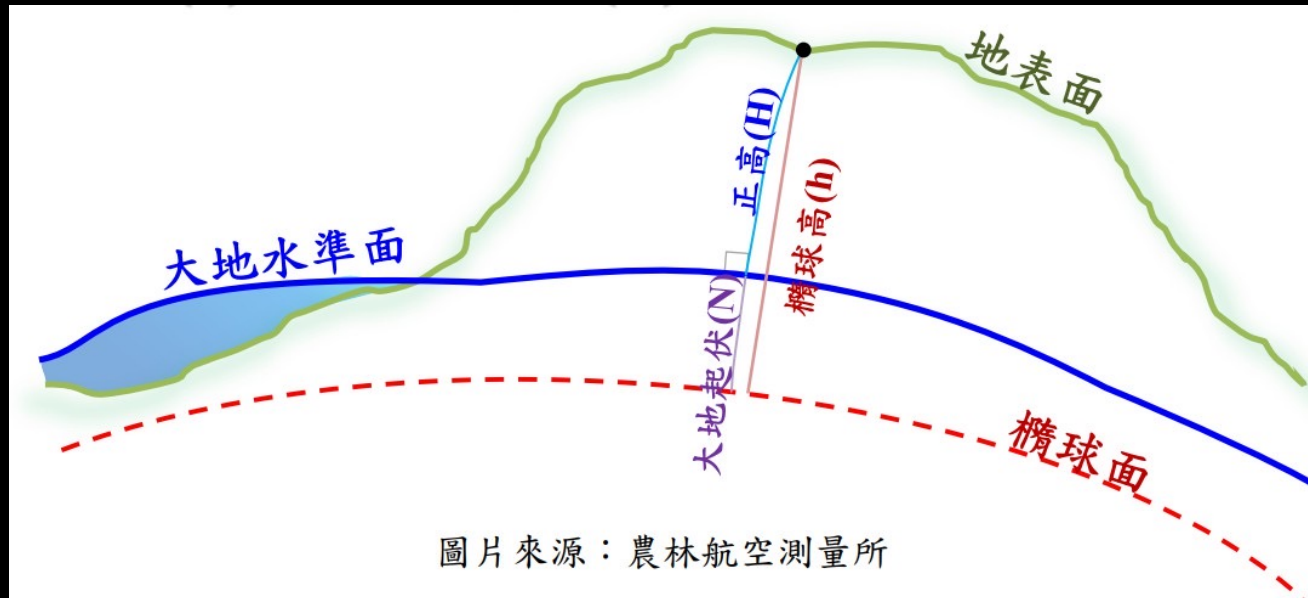


2. Reference Ellipsoid (參考橢球體)

Ellipsoid	Semi-major axis a [m]	Inverse flattening f^{-1}
Airy 1830	6,377,563.396	299.3249646
ANS	6,378,160.000	298.25
Bessel 1841	6,377,397.155	299.1528128
Clarke 1866	6,378,206.400	294.9786982
Clarke 1880	6,378,249.145	293.465
GRS80	6,378,137.000	298.257222101
International 1924	6,378,388.000	297.0
WGS84	6,378,137.000	298.257223563

- WGS84 (World Geodetic System 1984) is the most popular reference ellipsoid in the world.

3. Geoid (大地水準面)



- Ellipsoidal height (橢球高): the vertical distance from the reference ellipsoid to a measurement point.
- Orthometric height (正高): the vertical distance from the geoid (mean sea level) to a measurement point.
- Geoidal undulation (大地起伏): the separation between these two.

Geodetic Datum (大地基準)

- A set of parameters that defines the origin, orientation, scale, and reference surface of a coordinate system used to describe positions on or near the Earth.

Datum: Taiwan Datum 1967

Geodetic Datum used in Taiwan - onshore

Taiwan Datum 1967 is a geodetic datum first defined in 1967 and is suitable for use in Taiwan, Republic of China - onshore - Taiwan Island, Penghu (Pescadores) Islands. Taiwan Datum 1967 references the GRS 1967 Modified ellipsoid and the Greenwich prime meridian. Taiwan Datum 1967 origin is Fundamental point: Hu Tzu Shan. Latitude: 23°58'32.34"N, longitude: 120°58'25.975"E (of Greenwich). Taiwan Datum 1967 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from National Land Surveying and Mapping Center (NLSC), <http://www.nlsc.gov.tw> Adopted in 1980. TWD67 uses the GRS 1967 ellipsoid but with 1/f to exactly 2 decimal places.

Datum Details

DATUM NAME:	Taiwan Datum 1967
CODE:	1025
AREA OF USE:	Taiwan - onshore
SCOPE:	Geodetic survey, cadastre, topographic mapping, engineering survey.
TYPE:	geodetic
REALIZATION EPOCH:	1967
ORIGIN:	Fundamental point: Hu Tzu Shan. Latitude: 23°58'32.34"N, longitude: 120°58'25.975"E (of Greenwich).
ELLIPSOID:	GRS 1967 Modified
PRIME MERIDIAN:	Greenwich
APPLICABLE CRS-S:	The following CRS are based on this datum: [TWD67]

Datum: Taiwan Datum 1997

Geodetic Datum used in Taiwan

Taiwan Datum 1997 is a geodetic datum first defined in 1997 and is suitable for use in Taiwan, Republic of China - onshore and offshore - Taiwan Island, Penghu (Pescadores) Islands. Taiwan Datum 1997 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Taiwan Datum 1997 origin is ITRF94 at epoch 1997.0 Taiwan Datum 1997 is a geodetic datum for Geodetic survey, GIS, topographic mapping, engineering survey. It was defined by information from National Land Surveying and Mapping Center (NLSC), <http://www.nlsc.gov.tw> Adopted in 1998.

Datum Details

DATUM NAME:	Taiwan Datum 1997
CODE:	1026
AREA OF USE:	Taiwan
SCOPE:	Geodetic survey, GIS, topographic mapping, engineering survey.
TYPE:	geodetic
REALIZATION EPOCH:	1997
ORIGIN:	ITRF94 at epoch 1997.0
ELLIPSOID:	GRS 1980
PRIME MERIDIAN:	Greenwich
APPLICABLE CRS-S:	The following CRS are based on this datum: [TWD97] [TWD97] [TWD97]

國際地球參考框架（**International Terrestrial Reference Frame**，簡稱為**ITRF**）。ITRF 為利用全球**GNSS測站網之觀測資料成果推算所得之地心坐標系統**。

Primary Coordinate System in Taiwan

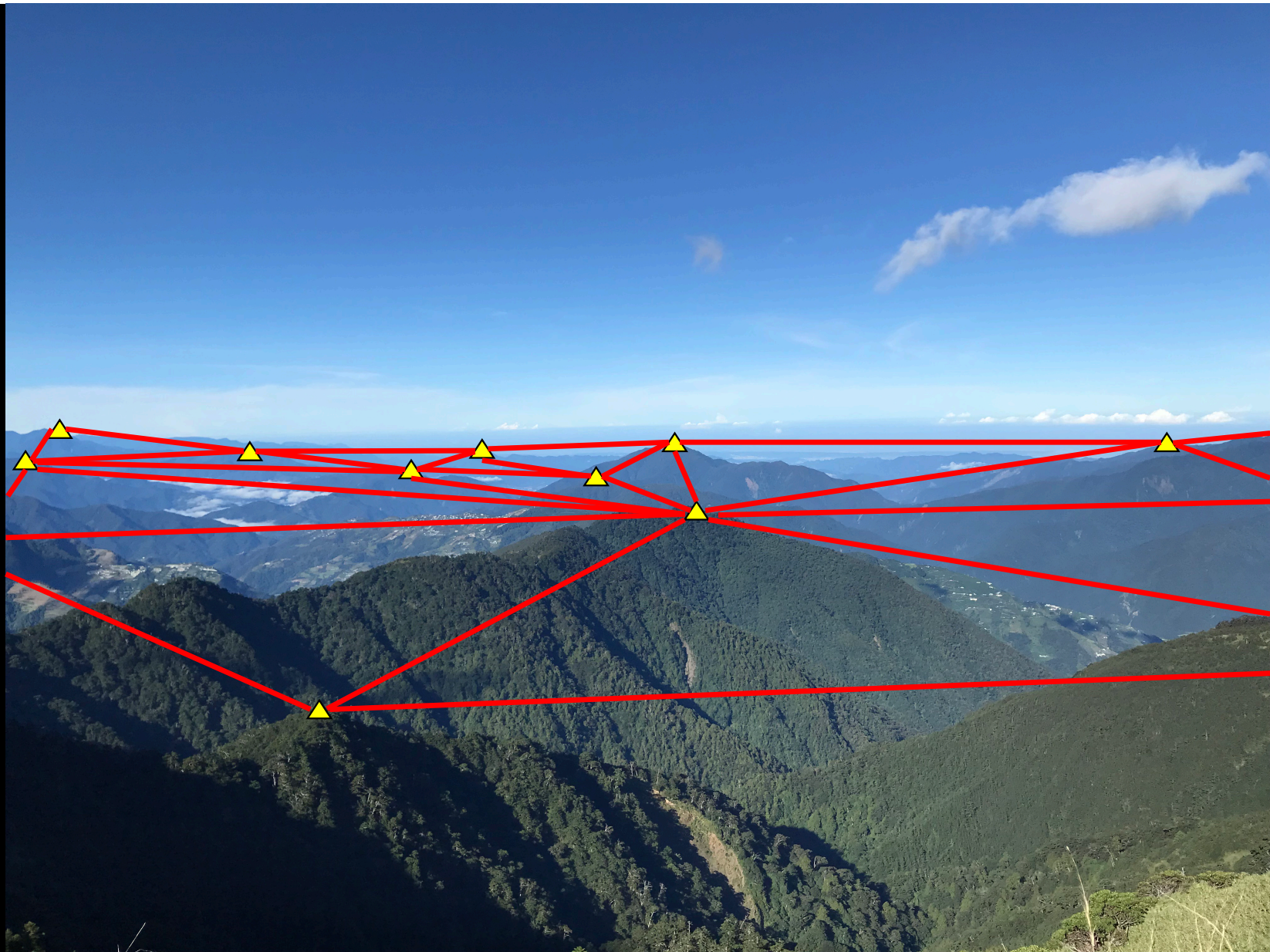
- Geographic coordinate system (GCS)
 - WGS84 – Latitude / Longitude
 - TWD97 – Latitude / Longitude
- Projected coordinate system (PCS)
 - Universal Transverse Mercator (UTM)
 - TW67 – TM2
 - TW97 – TM2

完整的投影坐標系統命名：Datum + Projection，例如：TWD 97- TM2

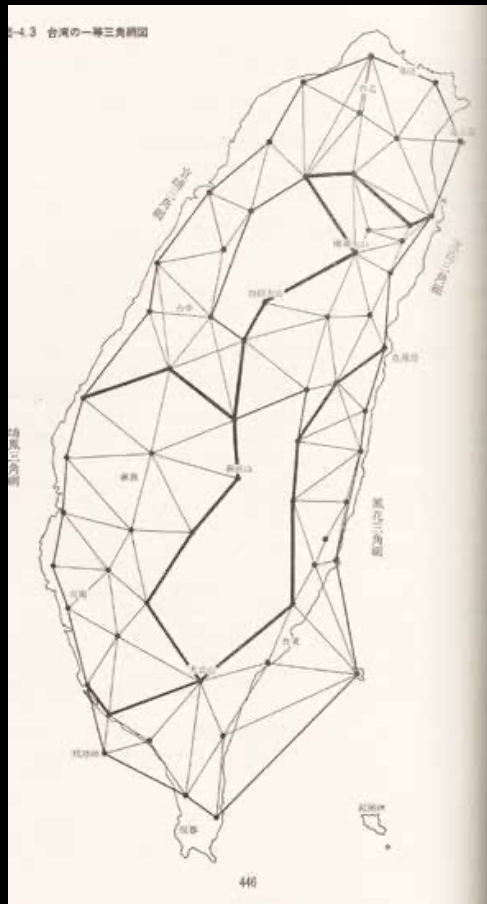
How to measure Earth's relief?





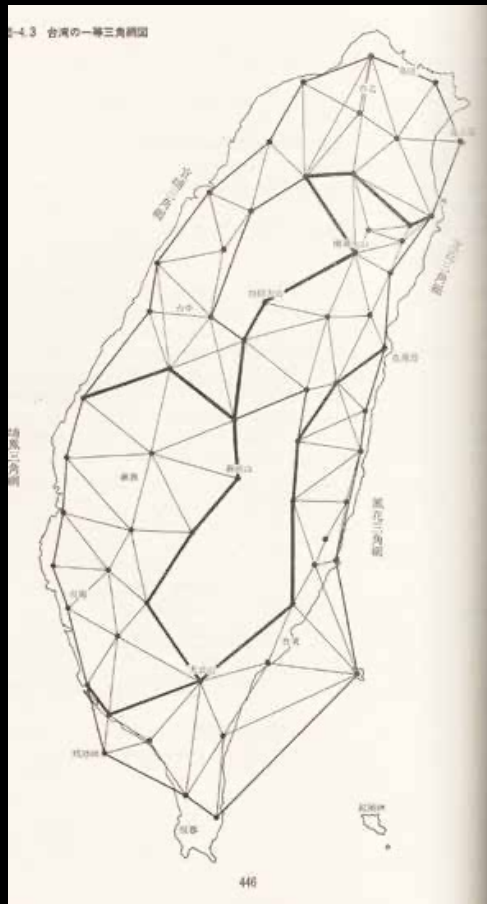


Triangulation (三角測量)

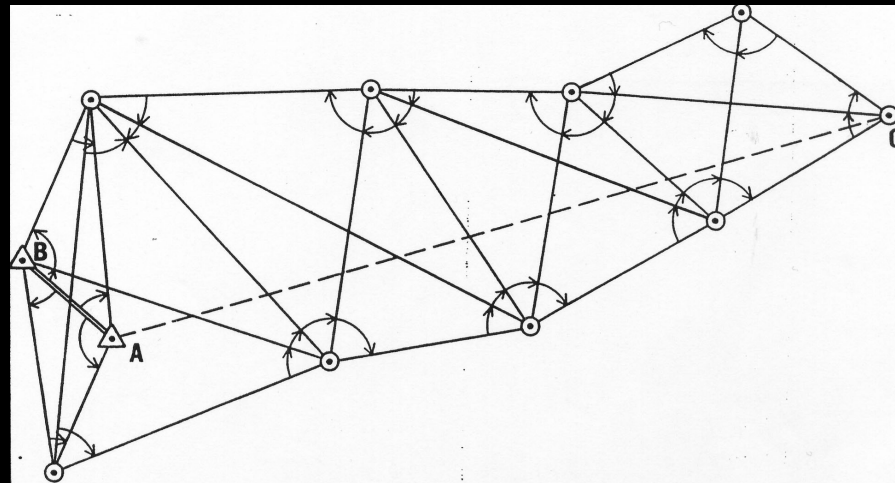


台灣一等三角點與三角網

Triangulation (三角測量)

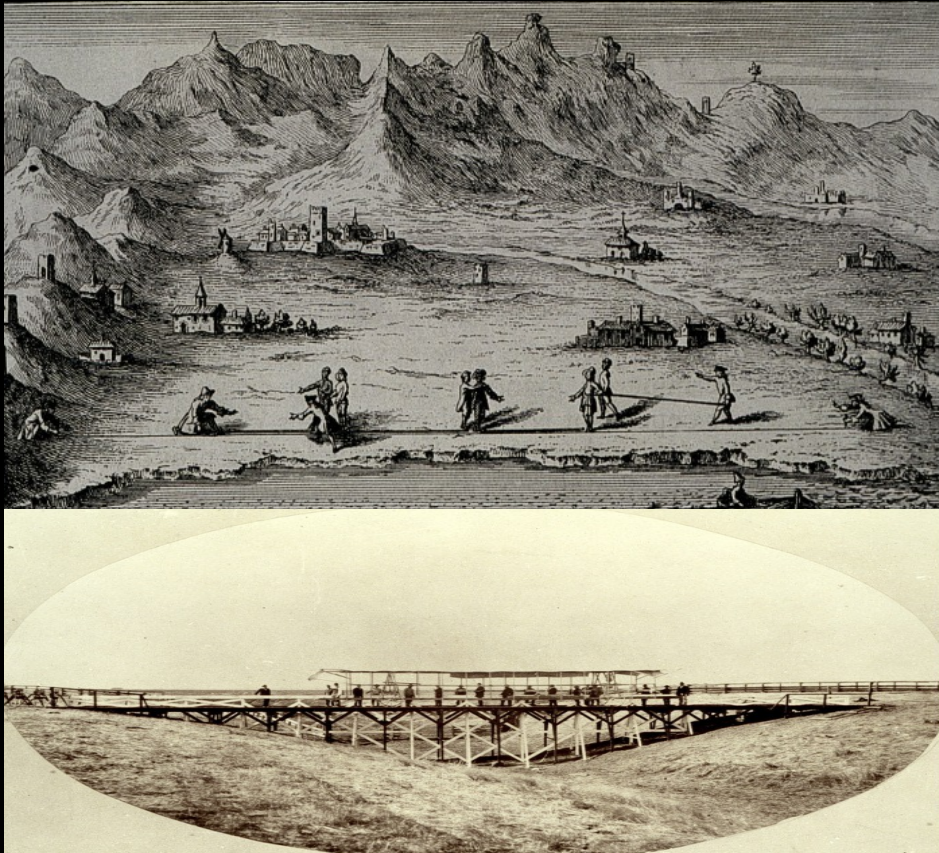


台灣一等三角點與三角網



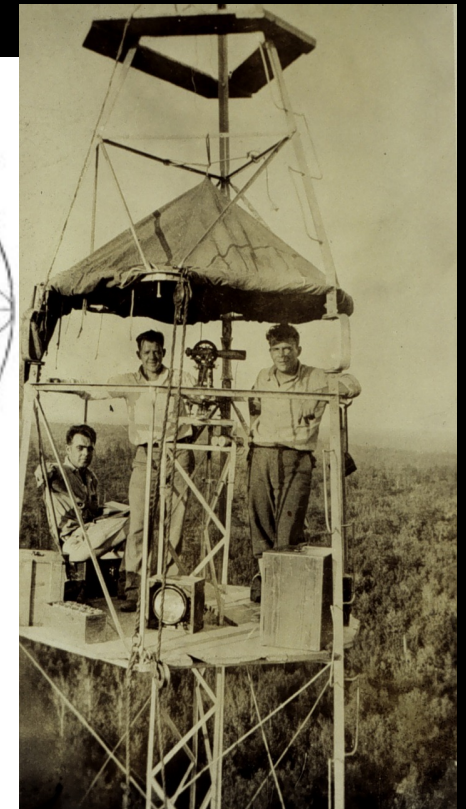
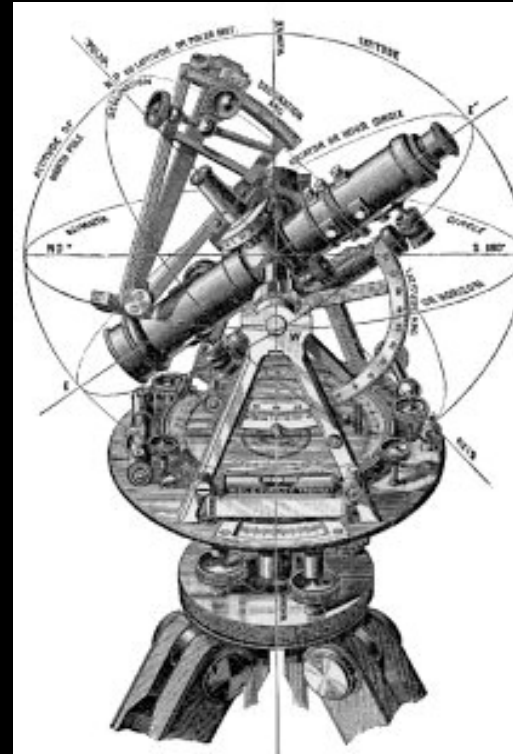
1. Determine position of A using astrogeodetic determination of Lat/Lon
2. Determine position of B by astronomic azimuth and baseline measurement

Triangulation (三角測量)



- Need to measure baseline length for scale

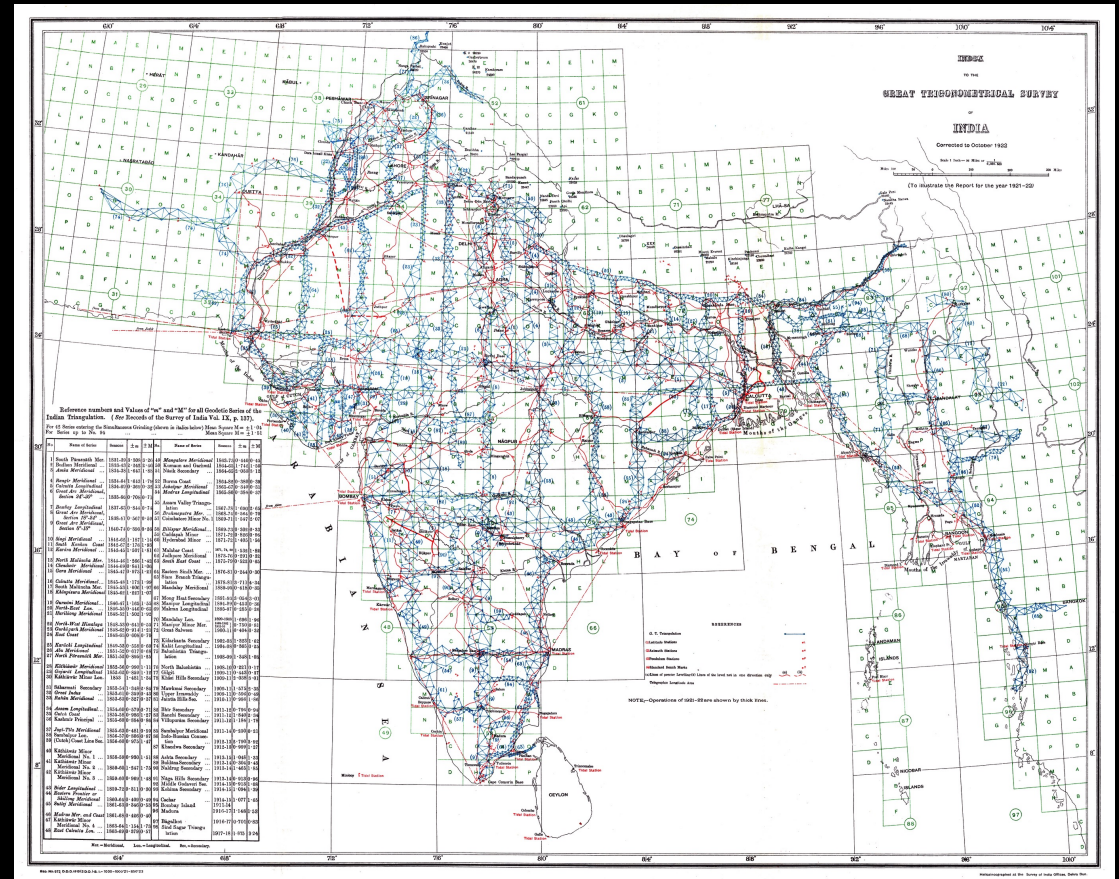
Theodolite (經緯儀)



- Depends on precise angle measurements using a theodolite
- Precision: $\sim 0.1 - 0.3''$ (1st order triangulation)

Great Trigonometrical Survey of India

- First accurate measurements of a section of an arc of longitude to refine estimates of Earth's curvature and ellipsoidal shape
- Determining the height of Mt. Everest
- Discovery of gravitational anomalies and geoid variation



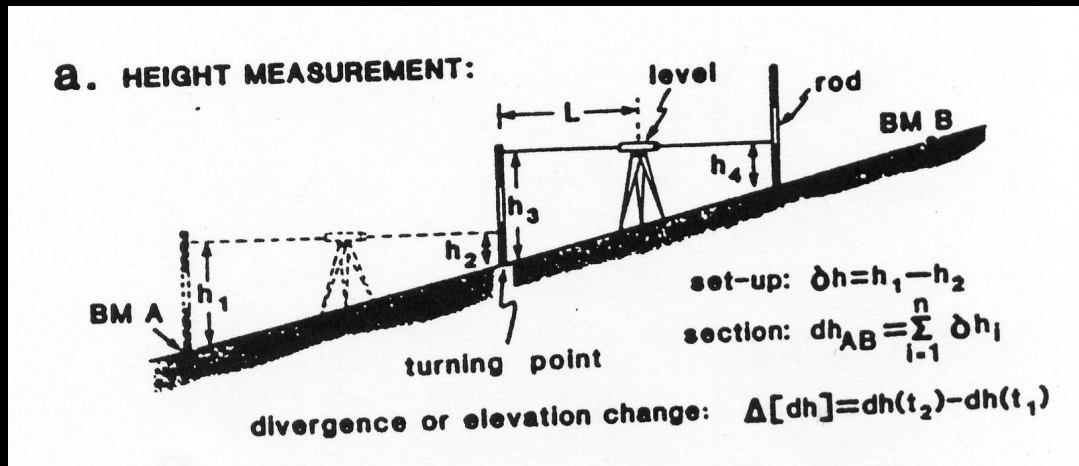
Trilateration (三邊測量)

- Measure distances within a network to locate points
- Multiple distance change measurements provide information about strain and displacements
- Electronic Distance Measuring (EDM)
- 2-color EDM
- Was used primarily in “permanent” installations at Long Valley and Parkfield, but also in several field experiments



Sprint Leveling (水準測量)

- Reading are made on two calibrated staffs ahead of and behind the instrument. The difference between readings is the difference in elevation between the points.

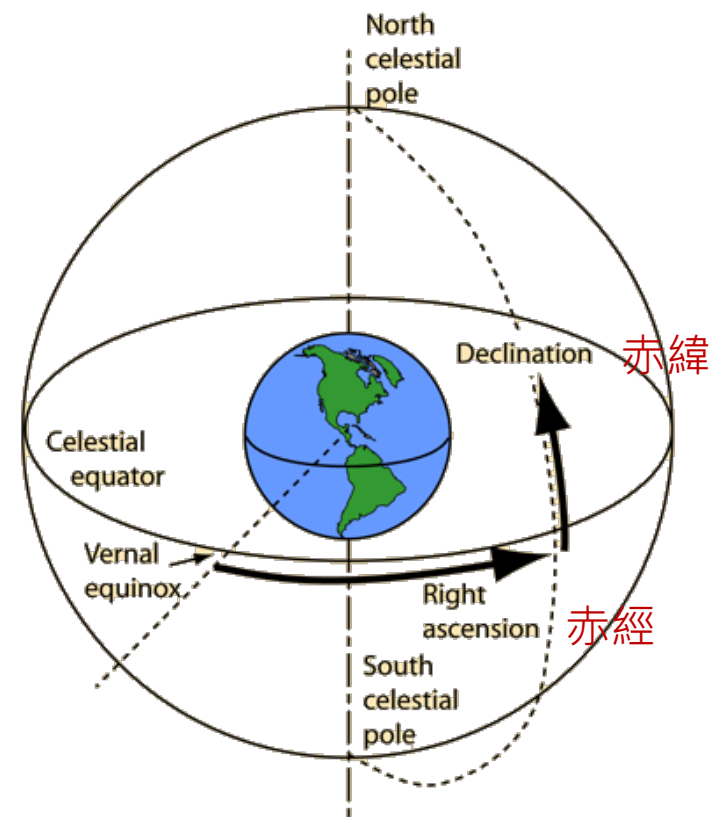
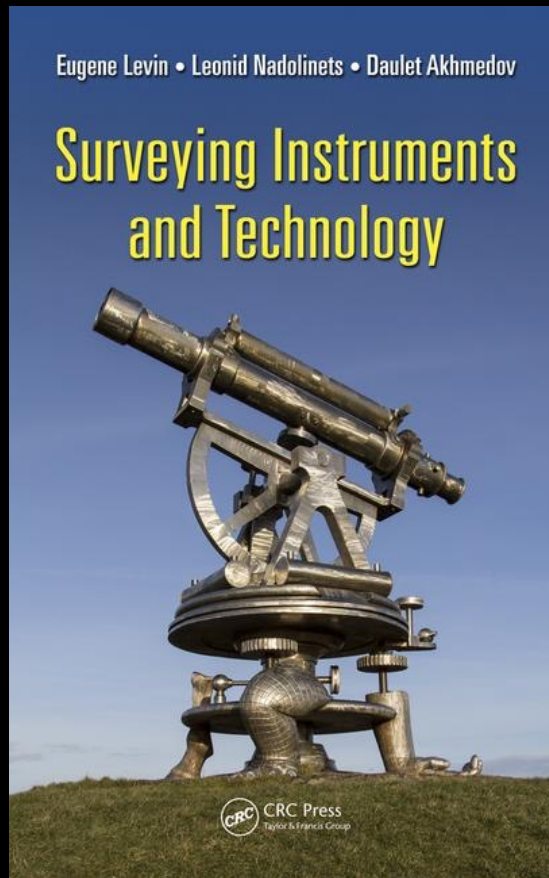


台灣水準原點

- 民國68年在基隆海門公園設置水準原點，此為臺灣首創之水準原點。舊有標石毀損，不復存在。
- 內政部於民國90年新設臺灣水準原點。後因配合基隆港東岸聯外道路工程興建，於民國103年將臺灣水準原點遷移至基隆國立海洋科技博物館對面的公園綠地，仍採雙水準原點設計，一為主點（點號：K997），屬地下點位，一為副點（點號：K996），屬地面點位。
- 水準原點之高程採用正高系統，以基隆平均海水面為參考依據，並據以訂定2001臺灣高程基準（TWVD2001）。



Astrogeodesy (天文測地學)



Terrestrial Geodetic Methods

- Importance:
 - Historic records going back as much as > 150 years
 - Complementary to space geodetic methods
 - Leveling, strainmeters, creepmeters, and EDM are still more precise than GPS/GNSS over short distances, under most circumstances
- Limitations
 - Do not allow measurements without line of sight and require in situ instrumentation
 - Often not very precise and laborious (expensive)
 - Error increase significantly with distance and/or time
 - Do not provide full 3D displacements, but limited information about displacement components and/or strain

Global Navigation Satellite System (全球導航衛星系統)

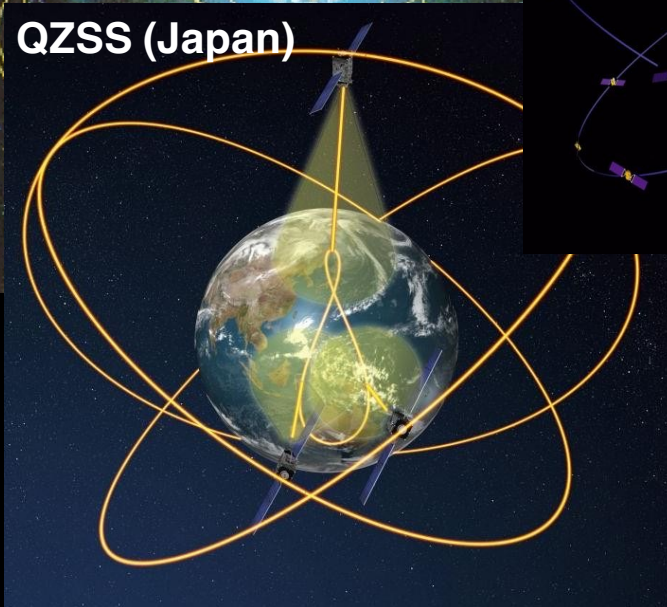
- Global Positioning System (GPS) (全球衛星定位系統)
- Continuous position fix
- Precise timing device
- Worldwide
- 3D coordinates
- ~10 m to mm accuracy in seconds to hours

GNSS: Not Only GPS

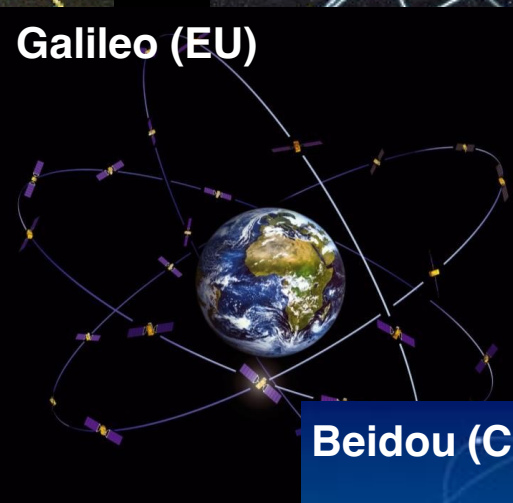
GPS (USA) the oldest GNSS



QZSS (Japan)



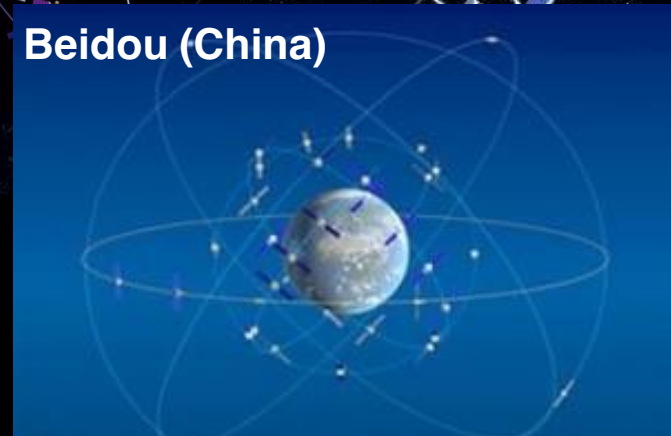
Galileo (EU)



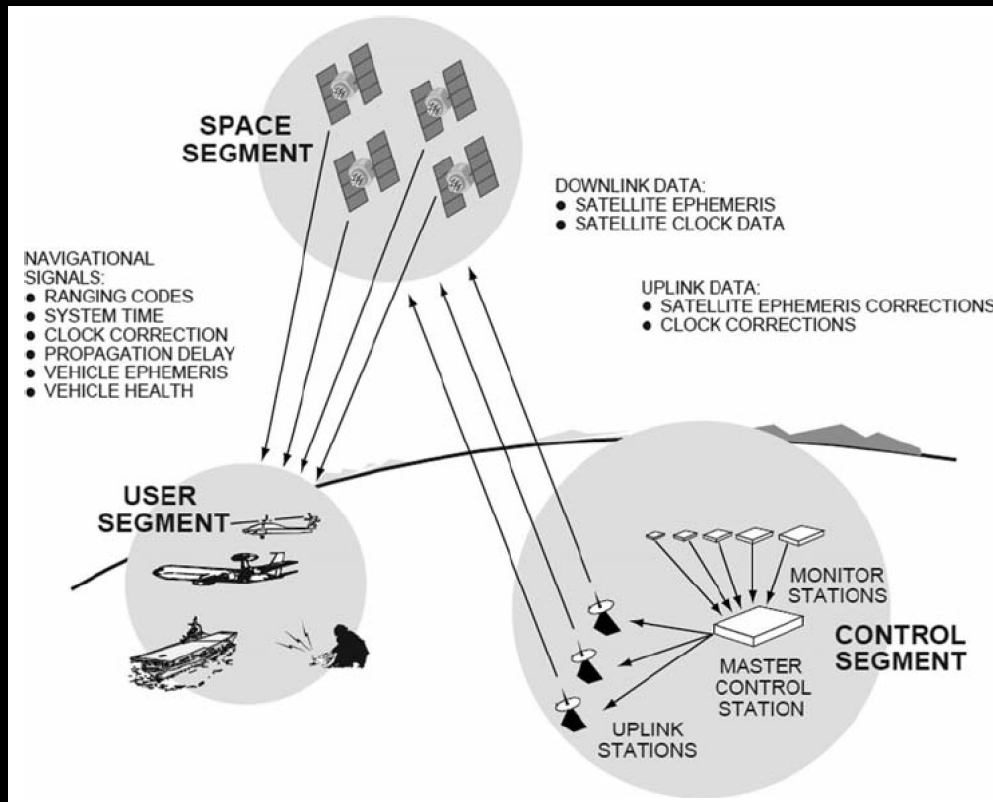
GLONASS (Russia)



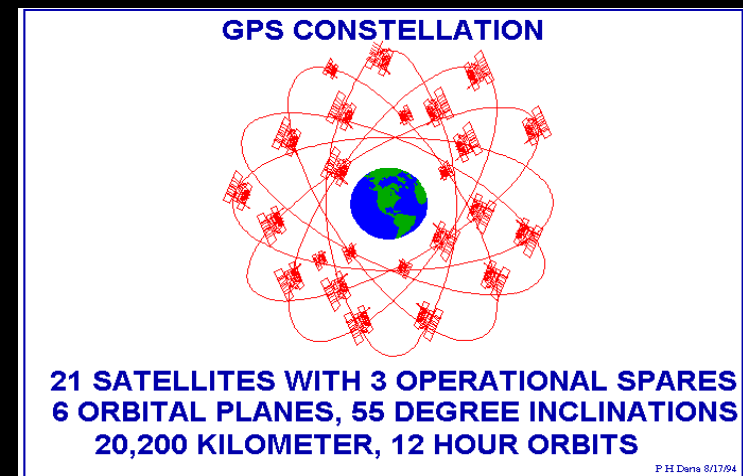
Beidou (China)



GPS System

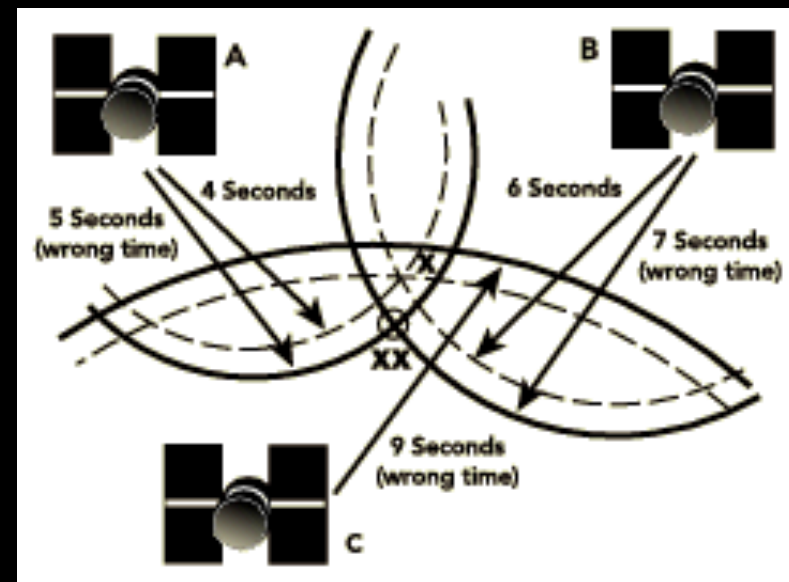
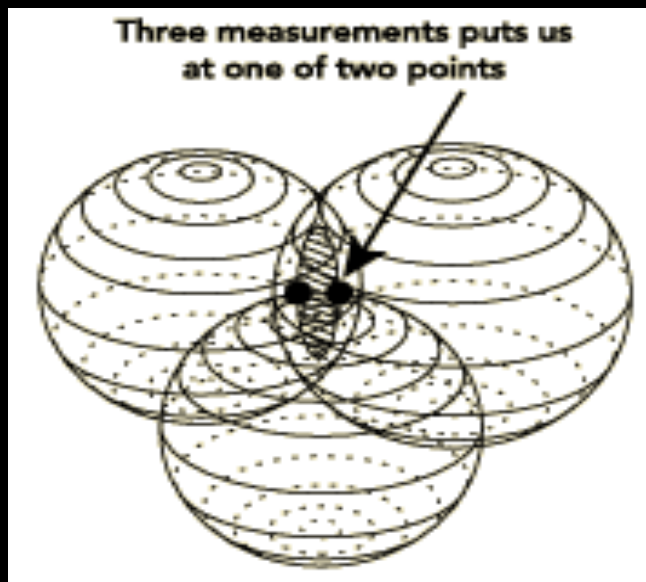


- Space Segment: Constellation of ≥ 24 GPS satellites (SV) broadcasting GPS signals
 - 6 orbital planes inclined at 55°
 - once/12 hrs on 20,200 km orbit
 - The current GPS constellation consists of 32 Block II/IIA/IIR/IIR-M satellites.



Positioning from Ranges

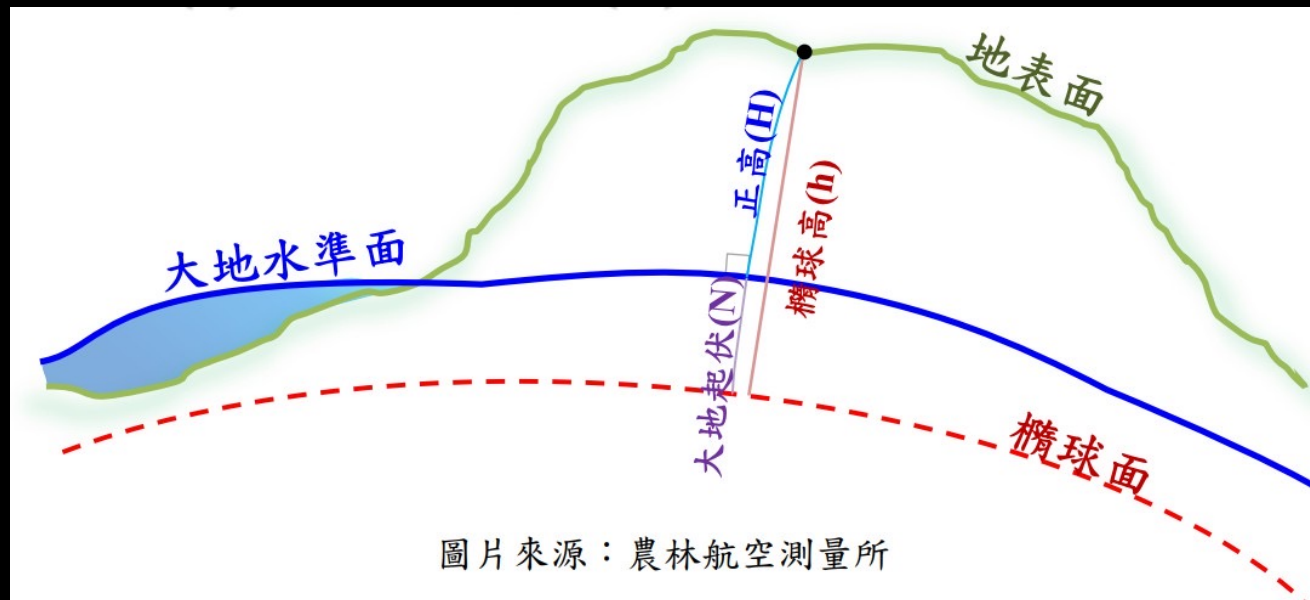
- Distance ranging to 3 satellites reduces the number of possible coordinates to two.
- A 4th satellite is needed to synchronize the imprecise receiver clock ($1\ \mu\text{s}$ $\sim$ 300 m error!)



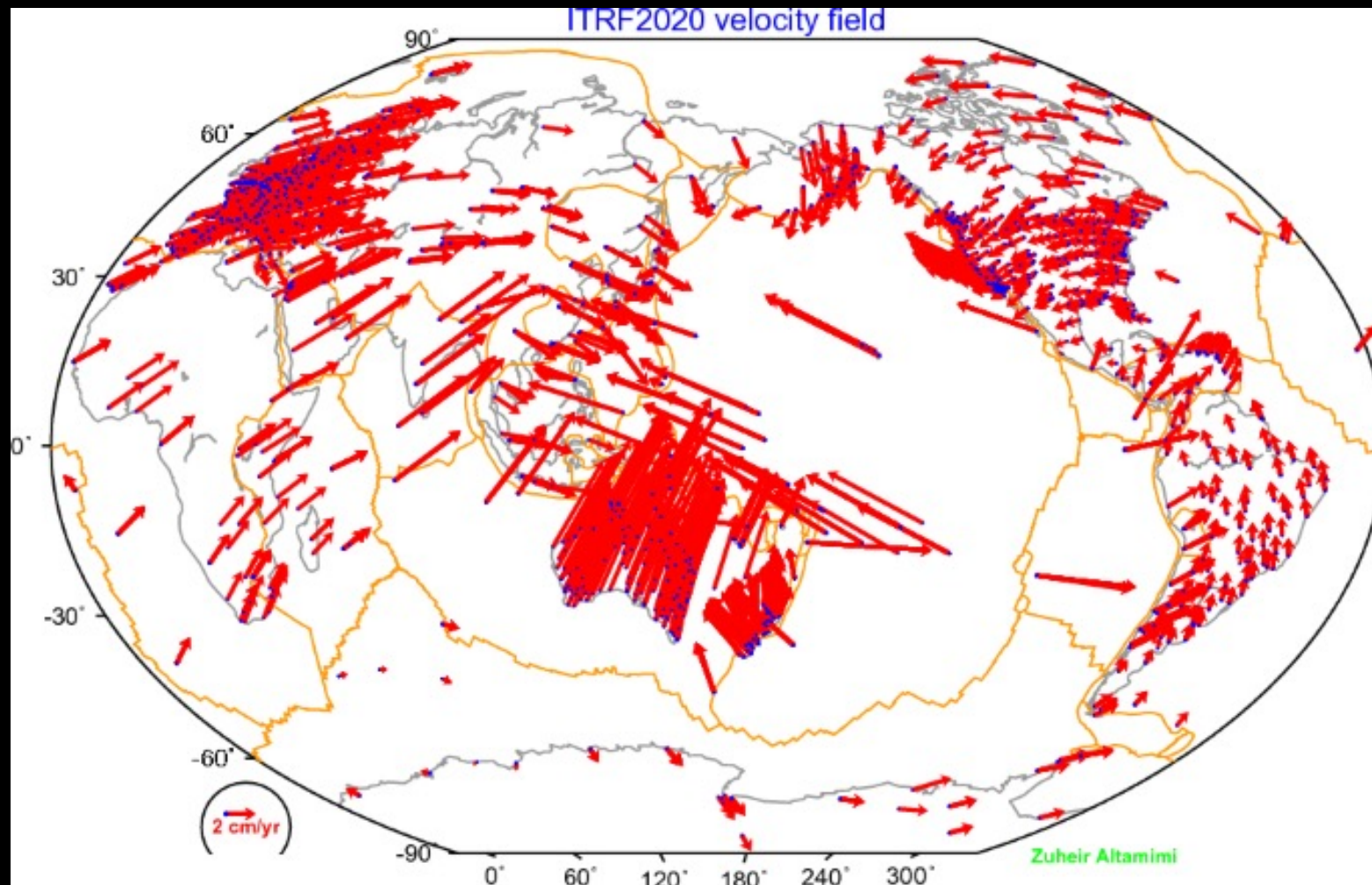
Reference Ellipsoid for GPS

- WGS84

- Semi-major axis (a) = equatorial radius (WGS-84: 6356752.3142 m)
- Semi-minor axis (b) = polar radius (WGS-84: 6378137.0 m)
- Flattening = $f = (a-b)/a$ (WGS-84: 1/298.25722356)
- First eccentricity squared = $e^2 = 2f-f^2$ (WGS-84: 0.00669437999013)

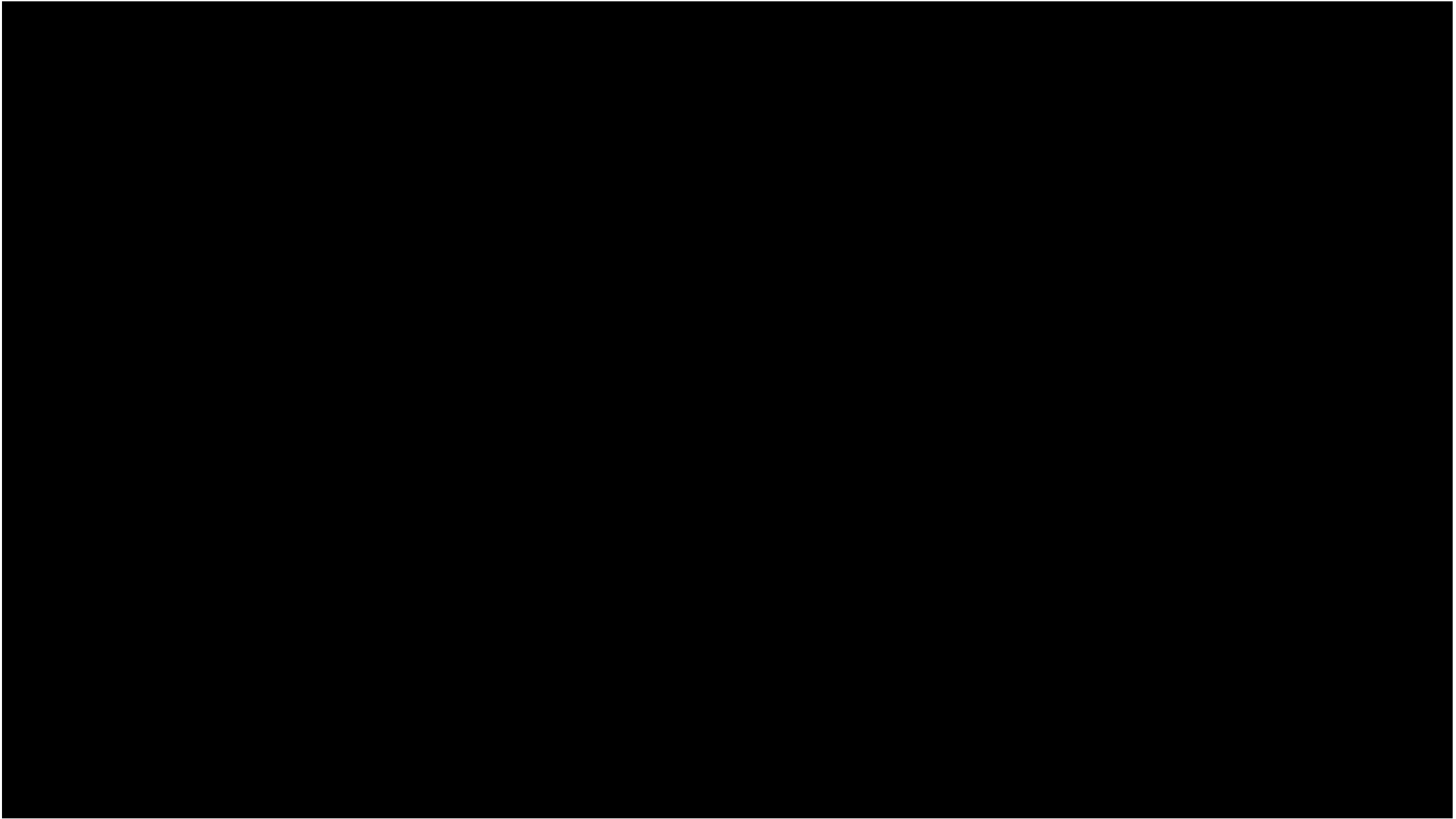


ITRF2020



Altamimi et al. (2023)

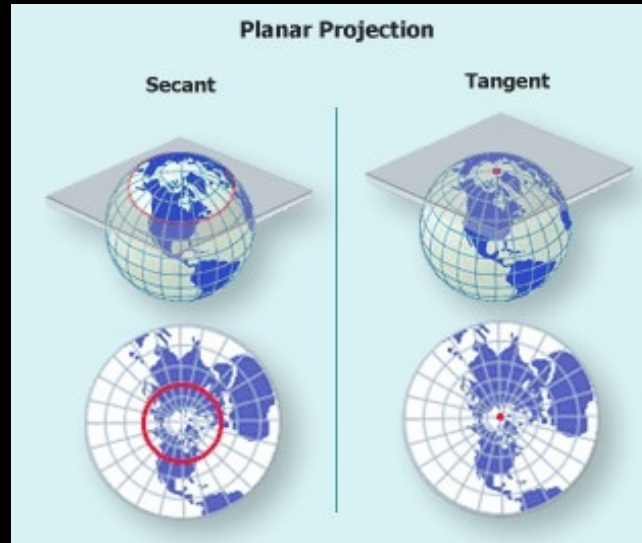
<https://link.springer.com/article/10.1007/s00190-023-01738-w>



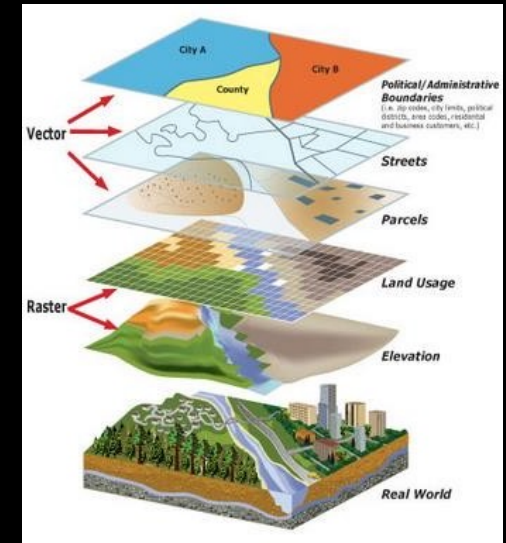
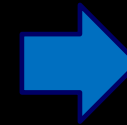
Key Concept



Geographic
Coordinate System



Map
Projection



Geographic
Data